Ryan Reece, Ph.D.

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EXPERIENCE

Sr. Staff Technical Lead and Manager, AI Cloud | Oct 2024 - present (2 yr 1 mo)
Staff Machine Learning Engineer | Oct 2022 - Oct 2024 (2 yr)
Tenstorrent, Santa Clara, CA

- Co-led the AI Cloud team as applications lead behind console.tenstorrent.com, Tenstorrent's public demo platform showcasing record-breaking DeepSeek chat and Wan video generation on Galaxy hardware; unveiled at the TT-Deploy launch event on May 1, 2026 alongside the GA of Blackhole Galaxy servers
- Launched Tenstorrent as a public inference provider on OpenRouter (Oct 2026), serving GLM 5.3 and Qwen3.8-27B; drove production validation of tool calling, structured outputs, and streaming
- Launched Tenstorrent's first external inference offering, NLP inference-as-a-service through EdenAI (June 2023)
- Architected and maintain chat.tenstorrent.com, an internal chat service used company-wide for both interactive use and programmatic workloads (up to 500k requests/day from a single internal customer), proxying both Tenstorrent-hosted models and 3rd party APIs (Anthropic, Google, OpenAI)
- Drove the strategic decision to adopt vLLM as an inference server, recognizing that Continuous Batching and PagedAttention were essential for realistic multi-user chat; brought up Llama3 70B and later DeepSeek R1 on Tenstorrent hardware in Cloud through close collaboration with the TT Metal kernel team
- Developed inference API demos across language (BERT, T5, Llama, DeepSeek), image (YOLO), and video (Wan) models; benchmarked performance across multiple generations of Tenstorrent accelerators
- Collaborated with full-stack engineers to build a Kubernetes-based cloud platform for push-button provisioning of Tenstorrent accelerators; led containerization of the AI software stack
- Co-led design and delivery of a QLoRA fine-tuning service for customer-provided data: wrote the core fine-tuner (single GPU → multi-GPU), decomposed product requirements into 50+ tracked deliverables for my team, and drove the selection of Metaflow as the orchestrator for managing fine-tuning jobs on Kubernetes
- Collaborated with the ML training team to bring up Tenstorrent's first multi-host training examples, training a single model across multiple servers with combined data and model parallelism
- Led bringup of slurm orchestration for multi-host training jobs; co-designed and implemented a slurm node-selection plugin for topology-aware (really graph-aware) job scheduling
- Driving early experiments in agentic applications running on Tenstorrent hardware, backed by OpenClaw, to demonstrate Galaxy's suitability for agentic workloads beyond chat inference

Machine Learning Engineer | Apr 2018 - Aug 2022 (4 yrs 4 mos)
Cerebras Systems, Sunnyvale, CA

- Developed end-to-end model references in both PyTorch and TensorFlow, including input data pipelines for Cerebras Wafer-Scale Engines; shipped to customers in the Cerebras Model Zoo with examples and detailed documentation
- Trained benchmark models for computer vision (ResNets) and NLP (GNMT, Transformer, Linformer, BERT, RoBERTa, GPT-2) on GPUs to establish accuracy and convergence baselines; the same implementations were run by the QA/Performance team on Cerebras CS systems; evaluated the impact of mixed precision, sequence-length bucketing, and activation sparsity on convergence
- Helped develop a new normalization layer, OnlineNorm, that uses streaming statistics to allow normalization of activations with small batch sizes [*NeurIPS 2019*]
- Directly engaged and supported customers (from both national labs and industry) in meetings and on-sites; including helping Pittsburgh Supercomputing Center (PSC) select among scientific proposals for using their dual CS-1 systems: Neocortex
- Triaged, explored, and tested customer-shared models; represented customer requirements to compiler engineers; debugged model and data pipeline issues for customers

- Co-authored a blog about Getting started with PyTorch BERT models on the Cerebras CS-2 System

Postdoctoral Research Fellow | Jul 2013 - Aug 2017 (4 yrs 2 mos)
Santa Cruz Institute for Particle Physics, The University of California, Santa Cruz, and
The European Council for Nuclear Research (CERN), Geneva, Switzerland

- 11 years (postdoc and Ph.D.) as a member of the ATLAS experiment, a 3000+ person collaboration looking for new physics in high energy proton-proton collisions at the Large Hadron Collider (LHC)
- Long involvement in codebase of more than 10 million lines of C++ and almost as many lines of Python
- Expert in petabyte data reduction (ATLAS ~10 PB/year), world-wide grid computing, and data visualization as a user and primary supporter of our group's 200-CPU computing cluster, accumulated more than 350k CPU-hours
- Co-led analysis groups in multiple searches for signals of supersymmetry and exotic decays, contributed to 6 research publications, and defended their approval
- 2015-17, full-time operations support for the data acquisition system (DAQ) and detector monitoring systems of the SCT (a tracking sub-detector in ATLAS)
- 2016-17, built more expertise in machine learning techniques, deep learning frameworks using Keras to build CNNs for particle classification, and another project using sklearn for anomaly detection by clustering with Gaussian Mixture Models

Graduate Researcher | Jun 2006 - Jul 2013 (7 yrs)
The University of Pennsylvania, Philadelphia, PA, and
The European Council for Nuclear Research (CERN), Geneva, Switzerland

- First summers as a student with Penn (2006-08) at CERN participating in the integration and commissioning of custom electronics for the Transition Radiation Tracker (TRT), the outermost sub-detector of the ATLAS tracker
- 2009-12, throughout most of the running of the LHC, rotated the on-call responsibility for the TRT DAQ
- Ph.D. research with the data from ATLAS focused on the identification of decays of tau leptons and their use in searches for new physics, a pattern recognition problem to identify a type of particle
- 2009-10, was the lead developer of the cut-based tau identification used with the first ATLAS data
- 2010-12, helped develop advanced tau identification using Boosted Decision Trees (BDTs) which superseded the above
- Knack for developing data analysis frameworks: *e.g.* pyframe has been used by several analyses in ATLAS
- The ATLAS and CMS experiments at the LHC discovered the long-sought-after Higgs boson, evidence of which was announced on July 4, 2012 [*Physics Letters B*, arxiv:1207.7214]

EDUCATION

- **Ph.D. Experimental Particle Physics**, The University of Pennsylvania (Philadelphia, PA), Jun 2006 - Jul 2013
  thesis: "A search for new physics in high-mass ditau events in the ATLAS detector"
- **B.S. Physics with Honors**, The University of Texas (Austin, TX), Aug 2003 - May 2006
  thesis: "Late pulsing in the Hamamatsu R1408 PMT used in the Sudbury Neutrino Observatory"

PUBLICATIONS

- Chiley, V. *et al.* (2019). Online normalization for training neural networks. *NeurIPS 2019*. [arxiv:1905.05894]
- Albertsson, K. *et al.* (2018). Machine learning in high energy physics community white paper. [arxiv:1807.02876]
- As a member of the ATLAS collaboration since June 1, 2008, I am an "author" of more than 800 publications (google scholar, inspire). I made substantial contributions to 14 publications in scientific journals and several more physics conference proceedings. My favorites:

  1. Search for supersymmetry in a final state containing two photons and missing transverse momentum in $\sqrt{s} = 13$ TeV pp collisions at the LHC using the ATLAS detector. *European Physical Journal C, 76*, 517 (2016). [arxiv:1606.09150]
  2. Identification and energy calibration of hadronically decaying tau leptons with the ATLAS experiment in pp collisions at $\sqrt{s} = 8$ TeV. *European Physical Journal C, 75*, 303 (2015). [arxiv:1412.7086]

3. A search for high-mass resonances decaying to $\tau^+\tau^-$ with the ATLAS detector. *Physics Letters B, 719,* 242-260 (2013). [arxiv:1210.6604]
4. Performance of the ATLAS detector using first collision data. *Journal of High Energy Physics, 9,* 56 (2010). [arxiv:1005.5254]

SKILLS

- **General:** deep learning (NLP and CV), statistical analysis, data visualization, data-driven modeling, anomaly detection, neural network classifiers, boosted decision trees, petabyte data reduction, object-oriented design, polymorphic interfaces, writing technical reports, working independently and in groups, presenting my ideas, graduate level physics and mathematics
- **Programming languages, fluent:** C/C++/STL (23+ years), Python (20+ years);
  experienced: Go, JavaScript, SQL; **Markup languages:** LaTeX, Markdown, html with css
- **ML / Data science software:** PyTorch, TensorFlow, HuggingFace, vLLM, LiteLLM, Metaflow, OpenClaw, Keras, matplotlib, numpy, scipy, scikit-learn, pandas, jupyter, ROOT, pyhf, RooStats, TMVA
- **General software:** Linux, bash, git, Slurm, Kubernetes, AWS (EC2, S3), Docker, Mathematica, QT, UML
- **Hobbies:** poker, philosophy, cycling, climbing

Last updated: Nov 10, 2026